EduAirControl · Technical Manual

SENA INDUSTRIA

EDUAIRCONTROL

Technical Manual

Architecture, components, data, API and maintenance

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Version 1.1

Version Control

Version	Date	Description
1.0	October 5, 2026	Initial creation of the technical manual.
1.1	October 7, 2026	Alignment with the current code: microservices architecture, API Gateway/OAuth2, RBAC roles, s

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1. Introduction

This technical manual documents the construction of EduAirControl, a platform for the management, monitoring and analysis of educational environments using environmental and IoT data. It is intended for developers, support staff, instructors and those responsible for maintaining the system.

2. Objective

To describe the architecture, technologies, code structure, persistence, services, security, device communication and maintenance recommendations for version 1.0 of EduAirControl.

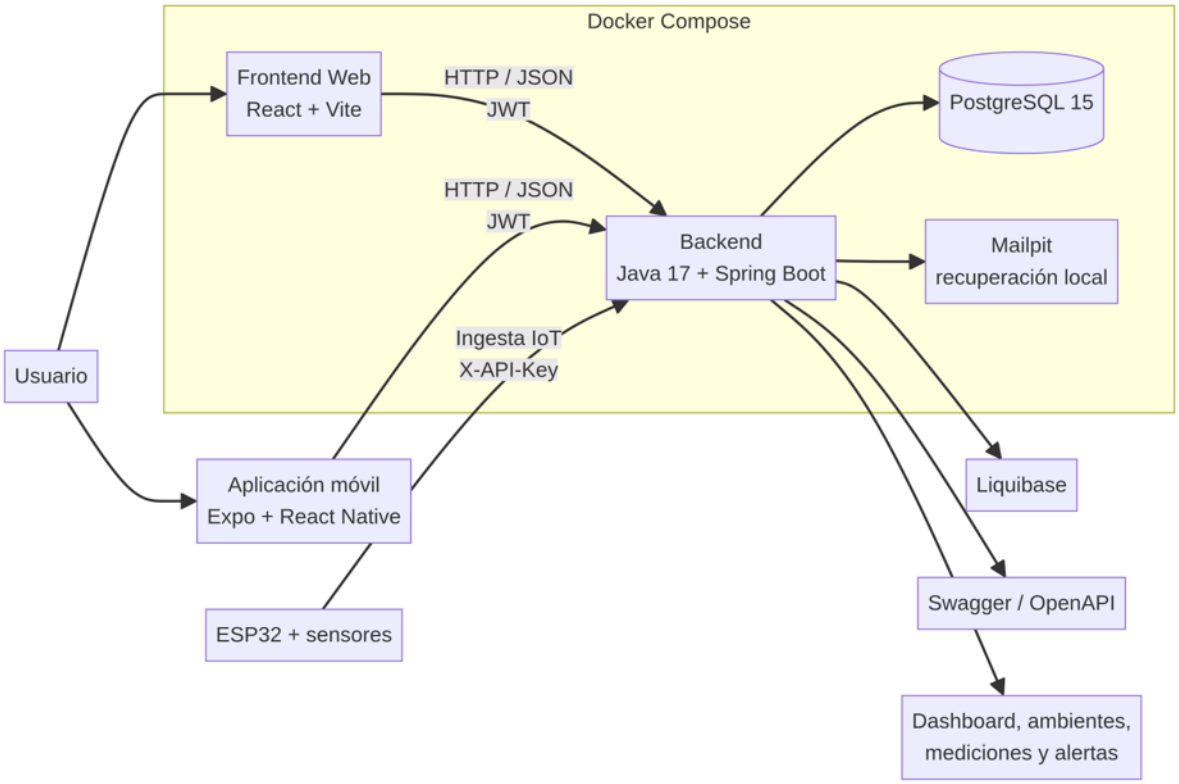
3. Scope

The document covers the web system, the backend, the PostgreSQL database, the Expo/React Native mobile application, the Docker services and the conceptual integration with the ESP32. The ESP32 files included in the delivery correspond to sensor tests; the final firmware ingestion must be confirmed before going into production.

4. Technical Description of the System

EduAirControl is organized as a client-server solution. The web frontend and the mobile application consume the backend over HTTP and JSON. The backend manages authentication, environments, sensors, measurements, analysis, alerts, preferences and notifications. PostgreSQL stores the information and Liquibase controls the evolution of the schema.

5. Architecture



EduAirControl is implemented as a microservices architecture with Spring Boot. Each service is an independent project with its own database (a dedicated schema per ADR-003) and follows a hexagonal architecture (ADR-009). The services are: api-gateway (port 8080), ms-security (8081), ms-classroom-management (3002), ms-environment-monitoring (3003), ms-sensor-management (3004) and ms-user-management (3007). The api-gateway acts as a BFF (Backend for Frontend) with OAuth2/OIDC (ADR-017).

5.1 Environmental Information Flow

1. The sensors capture variables of the educational environment.
2. The ESP32 processes the readings and, in the final integration, sends them over the network to the ingestion endpoint.
3. The backend validates the device's authentication, processes the reading and stores it in PostgreSQL.

4. The web frontend and the mobile application query the API and present measurements, statuses, analyses and alerts.

5.2 Components and Versions

Component	Technology / version	Function
Web frontend	React 19.2.0, Vite 8.0.1	Web interface and navigation
Backend	Java 17, Spring Boot 4.0.5	API, business logic and security
Persistence	PostgreSQL 15 + Spring Data JPA	Information storage
Migrations	Liquibase	Schema creation and versioning
Authentication	JWT 0.11.5	Sessions and authorization
API	Springdoc OpenAPI 2.5.0	Swagger documentation
Mobile	Expo 54.0.37, React Native 0.81.5	Android/iOS application
Containers	Docker Compose	Local orchestration
Local mail	Mailpit	Password recovery testing
Testing	JUnit/Spring Boot Testcontainers	Automated validation
API Gateway	Spring Cloud Gateway	Routing, authentication, rate limiting
Auth Server	Spring Authorization Server 7	OAuth2/OIDC, issuance of RS256 JWTs
Social Login	Google OAuth2, Facebook Login	Login with social networks
Roles	SUPER_ADMIN, ADMIN, USER, VIEWER	RBAC access control

6. Project Structure

```
EduAirControl/
   Web/
      Front-End/              # React + Vite
      backend/                # Java + Spring Boot
      docker-compose.yml      # local services
      docker-compose-jenkins.yml
   MOBILE/                    # Expo + React Native
   database/                  # Liquibase changelogs and models
   iot/esp32/                 # ESP32 sensor tests
   ms-monitoreo/              # analysis microservice under construction
   jenkins/                   # auxiliary Jenkins image
   Docs/                      # project documentation
```

Path	Responsibility
Web/Front-End/src	Screens, components, services, viewmodels, routes and internationalization
Web/backend/src/main/java	Identity, classrooms, sensors, monitoring, analysis and UX modules
Web/backend/src/main/resources	Properties and packaged Liquibase changelogs
database/	Data model separated by domain and migration files
MOBILE/.../src	Navigation, screens, services, BLE, storage and mobile styles
iot/esp32/	Diagnostic programs and DHT11/KY-038 tests

7. Backend Technical Modules

Module	Responsibilities	Highlighted elements
identity	Registration, login, JWT, recovery and profile	AuthController, ProfileController, User, Role
classrooms	Campuses, types and educational environments	CampusController, EnvironmentTypeController,
sensors	Devices, sensors, variables and ingestion	DeviceController, SensorController, IngestContro
monitoring	Measurements, dashboard, thresholds, alerts and notifications	DashboardController, NotificationController, Mea
analysis	Environmental analysis by period	AnalysisController, EnvironmentalAnalysis
ux	Favorites, preferences and notification states	FavoritesController, PreferencesController

Module	Responsibilities	Highlighted elements

8. Database

The database uses PostgreSQL 15 and four main domains. Hibernate is configured with `ddl-auto=validate`; therefore, Hibernate validates the schema and Liquibase is the source of truth for creating and evolving it.

Schema	Content
segurity	user, credential, role, permission, sessions, recovery, 2FA and audit
classrooms	campuses, environment_types, educational_environment
sensors	sensor, sensor_installation, variable, sensor_variable and related catalogs
monitoring	severity, variable_threshold, environment_measurement, environmental_analysis, analysis_result, environment_a
ux	user_preferences, favorites, classroom_ratings and searches

The changelogs are located in `Web/backend/src/main/resources/db/changelog` and also in `database/`. The `dev` context can load sample environments, sensors, measurements and alerts; an empty context leaves only the schema and the reference seed data.

9. API and Endpoints

The main API is published at `http://localhost:8080`. User operations use JWT; device ingestion uses `X-API-Key`. Errors are returned in JSON with the fields `status` and `message`.

Method	Route	Purpose
POST	/auth/login	Authenticate user and return JWT
POST	/auth/register	Register user
POST	/auth/forgot-password	Request password recovery
GET/PUT	/api/v1/profile	Retrieve or update profile
GET/POST/PATCH/DELETE	/api/v1/environments	Retrieve and manage environments
GET/POST/PATCH/DELETE	/api/v1/sensors	Retrieve and manage sensors
GET/POST/PUT/DELETE	/api/v1/devices	Manage IoT devices
POST	/ingest/v1/measurements	Receive device readings using X-API-Key
GET	/api/v1/dashboard/summary	Dashboard summary
GET	/api/v1/dashboard/series	Historical series
GET	/api/v1/notifications	Retrieve notifications
GET/POST/DELETE	/api/v1/favorites	Manage favorites
GET	/health	Check backend status

10. Security

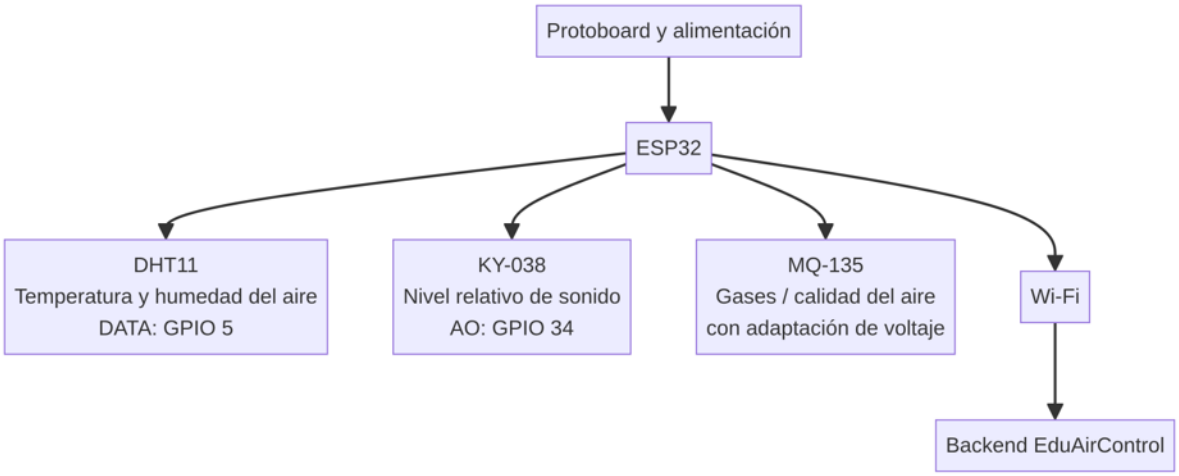
- JWT for user authentication and expiration control.
- Roles and protected routes to separate user and administrator operations.
- Social login (Google, Facebook) via the OAuth2 Authorization Code Flow.
- Super Admin: management of users, roles and institutions.
- Assignment of the ADMIN role to company administrators.
- `X-API-Key` device key for IoT ingestion; it is shown only once when the device is created or rotated.
- Sensitive variables through the environment: `JWT_SECRET`, `POSTGRES_PASSWORD`, `SMTP_PASSWORD` and administrator credentials.
- CORS configurable through `CORS_ALLOWED_ORIGINS`.
- Do not store real passwords, tokens or keys inside the repository or the manuals.

11. Web Frontend and Mobile Application

The web frontend uses React Router, HTTP services, forms, Zod/React Hook Form validations, i18next internationalization, Recharts charts and local preferences. The mobile application uses Expo, React Navigation, AsyncStorage, React Native BLE PLX, charts and the same authentication and query services.

Application	Main modules
Web	Landing page, authentication, dashboard, environments, management, favorites, profile, settings, notifications, a
Mobile	Authentication, dashboard, environments, favorites, notifications, profile, settings, device management and BLE

12. Hardware and IoT Integration



Component	Documented function / connection
ESP32	Controller and Wi-Fi communication
DHT11	Air temperature and humidity; tests with DATA on GPIO 5
KY-038	Relative sound level; AO output on GPIO 34 in tests
MQ-135	Gases and air quality; requires voltage adaptation before the ADC
Breadboard and jumpers	Prototype assembly
Battery	Portable power supply, voltage not yet confirmed

Technical warning: the MQ-135 must not be presented as a precision CO_2 meter, and its analog output must not be connected to an ESP32 GPIO without verifying the voltage level. The delivered files contain DHT11 and KY-038 tests at 115200 baud; no final firmware that sends readings to the API was found.

13. Maintenance

- Keep the frontend, backend, mobile app, changelogs and documentation synchronized.
- Update dependencies with review and testing; do not modify critical versions without validating Docker and Testcontainers.
- Back up the `postgres_data` volume before schema changes.
- Apply Liquibase instead of modifying tables manually.
- Rotate `JWT_SECRET` and device keys in real environments.
- Run `npm run lint`, `npm run build` and `mvn test` before delivering a version.
- Record changes to the firmware, pins, protocol and IoT sending frequency.

14. Glossary

Term	Definition
API	Interface that enables communication between applications.
ESP32	Microcontroller with Wi-Fi/Bluetooth connectivity.
JWT	Signed token used to authenticate user requests.
Liquibase	Tool for versioning database changes.
MQ-135	Gas and relative air-quality sensor; requires calibration.
MQTT	IoT messaging protocol considered in the architecture, not confirmed in the firmware.
X-API-Key	Authentication key used by IoT devices.
BLE	Bluetooth Low Energy, used by the mobile app to provision the ESP32.